

Global Technology & Internet Adoption Analysis (1990–2024): Insights from the World Bank’s World Development Indicators

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Objective / Problem Statement

Technology and internet access have become fundamental drivers of economic growth, education, innovation and social inclusion. However, the rate of technology adoption differs significantly across countries due to variations in infrastructure, economic development, affordability, and digital literacy.

The objective of this project was to analyze global technology and internet adoption trends using the World Bank's World Development Indicators (WDI) dataset. The project aimed to identify long-term trends, compare technology adoption across countries, evaluate the relationship between different communication technologies, and provide data-driven insights through an interactive Power BI dashboard.

Specifically, the analysis sought to answer the following questions:

1. How has global internet adoption evolved over time?
2. Which technology has experienced the greatest level of adoption?
3. Which countries lead in internet usage?
4. How does broadband adoption compare with mobile and fixed telephone subscriptions?
5. What insights can support future digital transformation policies and investments?

Dataset Description

The dataset used in this project was obtained from the World Bank World Development Indicators (WDI) database, one of the world's most comprehensive sources of global development statistics. The original dataset contains development indicators for over 200 countries and territories, covering multiple

sectors including education, health, economy, trade, environment, infrastructure, and technology.

For this project, the analysis focused exclusively on the Technology and Internet Adoption theme using the following four indicators:

- Individuals using the Internet (% of population)
- Mobile cellular subscriptions (per 100 people)
- Fixed telephone subscriptions (per 100 people)
- Fixed broadband subscriptions (per 100 people)

The dataset spans 1990 to 2024, allowing for a long-term analysis of global technology adoption patterns.

Data Cleaning Process

To prepare the dataset for analysis, several data transformation and cleaning procedures were carried out using Power Query in Power BI. The following steps were performed:

- I filtered the dataset to retain only the data from **1990-2024** and the four technology-related indicators relevant to the project.
- I removed aggregate regions and economic groups (such as North America, European Union, and Sub-Saharan Africa) to ensure that the analysis focused solely on individual countries.
- I unpivoted the yearly columns into a long-table structure, transforming multiple year columns into a single Year column and a corresponding Value column.
- I renamed columns to improve readability and consistency.
- I converted the Year column from decimal format to whole numbers.
- I converted the Value column to a numeric data type to enable accurate calculations.
- I cleaned and trimmed text fields to eliminate unnecessary spaces and inconsistencies.

- I verified that the dataset contained no duplicate records or missing values requiring removal.
- I retained zero values because they represent valid observations where technology adoption had not yet occurred rather than missing data.

Following the cleaning process, the final dataset contained **24,674** records across **214** countries, making it suitable for interactive analysis and visualization.

Analysis Methodology

The cleaned dataset was imported into Microsoft Power BI, where additional calculations and visualizations were created using DAX measures.

Key Performance Indicators (KPIs) developed:

- Total Countries
- Average Internet Users (%)
- Average Mobile Subscriptions
- Average Fixed Telephone Subscriptions
- Average Broadband Subscriptions

Visualizations created:

- A line chart illustrating global internet adoption trends from **1990 to 2024**.
- A bar chart identifying the **top ten** countries with the highest internet usage.
- A technology comparison chart comparing adoption levels across mobile subscriptions, internet users, fixed telephone subscriptions, and broadband subscriptions.
- A map visualization displaying the geographical distribution of internet usage around the world.
- Interactive country and year slicers enabling dynamic filtering of the dashboard.

These visualizations allowed for the identification of growth trends, regional disparities, technology adoption patterns, and country-level comparisons.

Key Insights

1. Mobile Technology Has Been the Primary Driver of Digital Adoption

The analysis shows that mobile cellular subscriptions significantly outperform other technology indicators, with an average of **59.0** subscriptions per 100 people, compared to **18.9** fixed telephone subscriptions and **11.5** fixed broadband subscriptions. The Technology Adoption Comparison further highlights this dominance, where mobile subscriptions recorded **414.62K**, far exceeding the values for fixed telephone and broadband technologies.

This indicates that many countries, particularly developing economies, bypassed traditional fixed-line infrastructure and adopted mobile technology as their primary means of communication and internet access.

2. Internet Adoption Experienced Rapid Growth After the Early 2000s

Global internet usage increased dramatically over the study period. The Internet Adoption Trend shows steady growth from approximately **3K** in **1990** to a peak of **43K** in **2024**, reflecting the rapid expansion of digital technologies worldwide.

Growth remained relatively slow during the **1990s**, accelerated significantly between **2000** and **2015**, and has continued to grow at a more stable pace in recent years as internet adoption matured across many developed economies.

3. Technology Adoption Remains Uneven Across Countries

The **Top 10** Countries by Internet Usage are dominated by high-income economies such as **Hong Kong SAR, Monaco, Switzerland, Sweden, Iceland, Denmark, Luxembourg, Germany**, and the **United Kingdom**.

This suggests that countries with stronger economies and more advanced digital infrastructure continue to lead global technology adoption, while many developing nations still experience lower levels of internet access.

The dashboard also reports an average global internet usage of **31.9%**, highlighting that a large portion of the world's population remains without reliable internet connectivity.

4. Broadband Adoption Continues to Lag Behind Mobile Connectivity

While mobile subscriptions continue to rise globally, fixed broadband adoption remains comparatively low. This trend suggests that mobile internet has become the preferred method of accessing digital services, particularly in emerging markets where fixed broadband infrastructure is limited or costly to deploy.

The findings emphasize the increasing reliance on mobile networks for education, commerce, financial inclusion, healthcare, and digital communication.

5. Global Digital Transformation Has Accelerated but the Digital Divide Persists

The overall trend demonstrates remarkable progress in global technology adoption over the past three decades. However, significant disparities remain between developed and developing regions.

While many countries have achieved widespread digital connectivity, others continue to face challenges such as inadequate infrastructure, affordability constraints, and limited digital literacy. Bridging these gaps remains essential to achieving inclusive and sustainable digital development worldwide.

Recommendations

1. Expand Investment in Mobile Broadband Infrastructure

Since mobile technology has become the primary gateway to internet access, governments and telecommunications providers should prioritize investments in 4G, 5G, and affordable mobile broadband services rather than relying solely on costly fixed-line infrastructure. This approach provides the fastest and most cost-effective path toward increasing internet accessibility, particularly in underserved communities.

2. Prioritize Low-Adoption Regions to Reduce the Digital Divide

The Global Internet Usage map highlights clear disparities in technology adoption across regions. Governments, development agencies, and private organizations should focus investments on countries with lower internet penetration, particularly across parts of **Africa**, **South America**, and developing regions of **Asia**,

through improved infrastructure, affordable devices, and connectivity programs. Targeted investments will help reduce inequalities in digital access while promoting economic growth and social inclusion.

3. Improve Internet Affordability and Digital Literacy

Although mobile subscriptions are relatively high, average internet usage remains considerably lower. This indicates that barriers to digital inclusion extend beyond infrastructure and include high internet costs, limited digital skills, and affordability challenges. Stakeholders should work together to:

- Reduce the cost of internet services.
- Promote digital literacy through education and training programs.
- Increase awareness of digital tools and online opportunities.

Addressing these challenges will encourage broader internet adoption and enable more meaningful participation in the digital economy.

Conclusion

This analysis demonstrates the remarkable growth of global technology and internet adoption over the past three decades, with mobile technology emerging as the primary driver of digital connectivity worldwide. While internet usage has expanded significantly since the early **2000s**, the findings also reveal persistent disparities in access between developed and developing countries. The dashboard highlights that **strengthening mobile broadband infrastructure, improving internet affordability, and expanding digital literacy initiatives** are critical to closing the global digital divide. By leveraging these insights, policymakers, governments, telecommunications providers, and development organizations can make informed decisions that promote inclusive digital transformation, drive economic development, and ensure that the benefits of technology are accessible to all.

References

World Bank. (2025). *World Development Indicators*.

<https://datatopics.worldbank.org/world-development-indicators/>